



To Whom It May Concern,

As a Lecturer at the College of Engineering Science and Technology, Shanghai Ocean University, over the past five years, I have led multiple municipal-funded research projects on marine robotics within the college. With close supervision of Mr. Xiong Yang in his coursework and research, I am writing to wholeheartedly recommend him for admission to your graduate program. I have been deeply impressed by his exceptional academic intuition, strong proficiency in algorithm implementation, and outstanding research output.

Yang first reached out to me in his sophomore year via a senior student in my lab. Driven by his deep interest in integrating artificial intelligence with robotic systems, he joined my research group and quickly adapted to the group's research workflow. Despite still being an undergraduate at that time, Yang demonstrated remarkably mature research skills, swiftly mastering literature review, experimental design, and quantitative data analysis. He initially contributed as a co-author to our paper in the Journal of Marine Science and Engineering (JMSE, SCI Q2 journal).

Fueled by his extraordinary academic enthusiasm, Yang later launched and led an intricate independent research project. Most notably, he published a first-author paper in JMSE entitled "Spatiotemporal Meta-Reinforcement Learning for Multi-USV Adversarial Games Using a Hybrid GAT-Transformer". He devised the novel Adv-TransAC framework to decouple spatial and temporal reasoning: Graph Attention Networks (GAT) handle real-time tactical formation planning, while Transformers conduct temporal intention inference. His algorithm outperformed cutting-edge MARL benchmark models by a notable margin in terms of task completion rate and policy robustness. In high-fidelity adversarial simulation environments, this algorithm also allows USV swarms to execute sophisticated, proactive collaborative strategies. The ability to independently complete such graduate-level algorithm design is solid proof of his outstanding research potential. His collaborative spirit is also evident in his co-authorship of numerous other SCI-indexed papers in journals including Sensors and Photonics.

Aware that solid theoretical groundwork underpins advanced practical research, Yang voluntarily took my elective course, Machine Learning and Its Applications. He did not merely complete coursework requirements; instead, he approached the subject with great diligence, regularly initiating in-depth discussions on algorithm optimization, and earned a perfect score in this course. Additionally, he can independently draft high-standard SCI manuscripts, which reflects his strong command of academic English writing.

To sum up, Xiong Yang stands out as an exceptional undergraduate who already boasts the research mindset and output consistency of an experienced researcher. He ranks among the most gifted students I have supervised throughout my academic career. He excels at applying state-of-the-art algorithms to tackle intricate engineering challenges. I am fully confident he is well-equipped to thrive amid the rigorous English-language coursework of your master's program, and I offer my unreserved highest recommendation for his admission.

Sincerely,

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